

Weikai Lin

CONTACT INFORMATION

Ph.D. Student
University of Rochester
Wegmans Hall
Rochester, NY 14627

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<https://linwk20.github.io/>
[Google Scholar](#) | [LinkedIn](#)

RESEARCH INTERESTS

I work at the intersection of computer graphics, computer architecture, and imaging systems. My research bridges the gap between hardware and software through system-level co-optimization. Current themes:

- **AR/VR neural rendering and acceleration:** neural rendering algorithms for AR/VR and their hardware accelerators. [CVPR'26, SIGGRAPH Asia'25, ASPLOS'25a, ASPLOS'25b, ISCA'25, HotMobile'25, TACO'24]
- **Imaging system design:** co-design of sensors and optics with downstream vision models. [DAC'26, DAC'25, WACV'25]

EDUCATION

University of Rochester, Rochester, NY, USA
Ph.D., Computer Science
M.Sc., Computer Science
Advisor: [Yuhao Zhu](#)

Aug 2023 – May 2028 (expected)
Aug 2023 – May 2025

Peking University, Beijing, China
M.Sc., Intelligence Science and Technology
Advisor: Dingsheng Luo

Sep 2020 – Jun 2023

Tsinghua University, Beijing, China
B.E., Electronic Engineering
Advisor: Li Su

Aug 2016 – Jun 2020

HONORS AND AWARDS

- 2026, ACM/IEEE DAC Young Fellow
- 2026, ACM ASPLOS Travel Grant
- 2025, ACM ASPLOS Best Paper Award
“MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering” (first author)
- 2025, OpenAI Research Grant (\$5K)
- 2025, ACM ASPLOS Travel Grant
- 2022, Baosteel’s Outstanding Student Award for HMT Students
- 2022, First Prize, Peking University Challenge Cup
- 2021, First Class Scholarship for Taiwan Master’s Students

PROFESSIONAL POSITIONS

Pixocial AI Lab, Bellevue, WA, USA
Applied Research Intern
Mentor: [Haoxiang Li](#)
Topics: Physics-based video diffusion and dynamic hair rendering. [\[project page\]](#)

May 2025 – Aug 2025

University of Rochester, Rochester, NY, USA
Graduate Researcher
Teaching Assistant

Aug 2023 – present
Fall 2024, Spring 2025, Fall 2025

Advanced Micro Devices, Inc. (AMD), Beijing, China
 Research Intern Jul 2022 – Feb 2023
 Mentor: Fuwei Yang.
 Topics: SLAM and DNN evaluations on AMD GPUs. Participated in a [ROCm](#) release.

Peking University, Beijing, China
 Graduate Researcher Sep 2020 – Jun 2023
 Teaching Assistant 2021 – 2022
 Topics: Robotics, robot learning, and autonomous driving.

Chinese Academy of Sciences (CAS), Beijing, China
 Engineer Intern
 Topics: HLS-based FPGA wireless communication system development. 2021 – 2022

Tsinghua University, Beijing, China
 Undergraduate Researcher Sep 2019 – Jun 2020
 Mentor: Li Su.
 Topics: FPGA-based wireless codec system development.

Cambricon Technologies Co., Ltd., Beijing, China
 Research Intern Jul 2019 – Aug 2019
 Mentor: Miao Li.
 Topics: Neural network pruning.

PUBLICATIONS

* denotes equal contribution.

- Ruofan Xing, Arman Akbari, [Weikai Lin](#), Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
[“HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training”](#)
 63rd ACM/IEEE Design Automation Conference (**DAC 2026**)
- He Zhu*, Xiaotong Huang*, Zihan Liu, [Weikai Lin](#), Xiaohong Liu, Zhezhi He, Jingwen Leng, Minyi Guo, Yu Feng
[“SeeLe: A Unified Acceleration Framework for Real-Time Gaussian Splatting on Mobile Devices”](#)
 IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)
CVPR 2026 Highlight
- [Weikai Lin](#), Sushant Kondguli, Carll Marshall, Yuhao Zhu
[“PowerGS: Display-Rendering Power Co-Optimization for Neural Rendering in Power-Constrained XR Systems”](#)
 SIGGRAPH Asia 2025 Conference Papers (**SIGGRAPH Asia 2025**)
- [Weikai Lin](#)*, Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
[“SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision”](#)
 62nd ACM/IEEE Design Automation Conference (**DAC 2025**)
- [Weikai Lin](#)*, Yu Feng*, Yuhao Zhu
[“MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering”](#)
 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2025**)
ASPLOS 2025 Best Paper Award
- Yu Feng*, Zheng Liu*, [Weikai Lin](#)*, Zihan Liu, Jingwen Leng, Minyi Guo, Zhezhi He, Jieru Zhao, Yuhao Zhu
[“StreamGrid: Streaming Point Cloud Analytics via Compulsory Splitting and Deterministic Termination”](#)
 30th ACM International Conference on Architectural Support for Programming Languages

and Operating Systems (ASPLOS 2025)

- Yu Feng, [Weikai Lin](#), Yuge Cheng, Zihan Liu, Jingwen Leng, Minyi Guo, Chen Chen, Shixuan Sun, Yuhao Zhu
[“Lumina: Real-Time Mobile Neural Rendering by Exploiting Computational Redundancy”](#)
52nd IEEE/ACM International Symposium on Computer Architecture (ISCA 2025)
- Adith Boloor, [Weikai Lin](#), Tianrui Ma, Yu Feng, Yuhao Zhu, Xuan Zhang
[“Private-Eye: In-Sensor Privacy Preservation Through Optical Feature Separation”](#)
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025)
- Nan Wu*, [Weikai Lin*](#), Ruizhi Cheng*, Bo Chen, Yuhao Zhu, Klara Nahrstedt, Bo Han
[“Advancing Immersive Content Delivery with Dynamic 3D Gaussian Splatting”](#)
26th International Workshop on Mobile Computing Systems and Applications (HotMobile 2025)
- Yu Feng*, [Weikai Lin*](#), Zihan Liu, Jingwen Leng, Minyi Guo, Han Zhao, Xiaofeng Hou, Jieru Zhao, Yuhao Zhu
[“Potamoi: Accelerating Neural Rendering via a Unified Streaming Architecture”](#)
ACM Transactions on Architecture and Code Optimization (TACO 2024)
- Wenfei Hu, [Weikai Lin](#), Hongyu Fang, Yi Wang, Dingsheng Luo
[“OW3Det: Toward Open-World 3D Object Detection for Autonomous Driving”](#)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024)
- Wenfei Hu, [Weikai Lin](#), Hongyu Fang, Yi Wang, Dingsheng Luo
[“Learning Clear Class Separation for Open-Set 3D Detector in Autonomous Vehicle Via Selective Forgetting”](#)
32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2023)
- Weiming Qu, Tianlin Liu, [Weikai Lin](#), Dingsheng Luo
[“A Review of Robot Learning”](#)
Acta Scientiarum Naturalium Universitatis Pekinensis, 59(6):1069–1086, 2023. DOI: 10.13209/j.0479-8023.2023.086
- Shuai Fang, Yaoyao Wei, [Weikai Lin](#), Jianan Zhang, Tianlin Liu, Dingsheng Luo
[“Approaching Sound Object with Sensorimotor Coordination when Sensors Partially Damaged”](#)
IEEE International Conference on Development and Learning (ICDL 2021)
- Yaoyao Wei, Shuai Fang, [Weikai Lin](#), Jianan Zhang, Dingsheng Luo
[“Acquiring Robot Navigation Skill with Knowledge Learned from Demonstration”](#)
IEEE International Conference on Development and Learning (ICDL 2021)
- Tianyu Hu, [Weikai Lin](#), Weizhi Zhang, Jing Ma, Song Wang
[“MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents”](#)
arXiv preprint arXiv:2605.00356, 2026
- Bo Yan, [Weikai Lin](#), Yada Zhu, Song Wang
[“SafeDream: Safety World Model for Proactive Early Jailbreak Detection”](#)
arXiv preprint arXiv:2604.16824, 2026
- [Weikai Lin*](#), Tianrui Ma*, Adith Boloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
[“SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision”](#)
CVPR 2026 On-Sensor Vision (OSV) Workshop [[link](#)]
- Ruofan Xing, Arman Akbari, [Weikai Lin](#), Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang

PREPRINTS AND WORKSHOPS

“HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training”
CVPR 2026 On-Sensor Vision (OSV) Workshop [\[link\]](#)

- Han Yan, Yifei Zou, [Weikai Lin](#), Xuan Zhang, Yuhao Zhu
“Do Energy-Harvesting Pixels Reduce Carbon Cost of Visual Computing Systems?”
HotEthics 2026 [\[link\]](#)
- [Weikai Lin](#), Haoxiang Li, Yuhao Zhu
“ControlHair: Physically-based Video Diffusion for Controllable Dynamic Hair Rendering”
arXiv preprint arXiv:2509.21541, 2025
- [Weikai Lin](#), Yu Feng, Yuhao Zhu
“RTGS: Enabling Real-Time Gaussian Splatting on Mobile Devices Using Efficiency-Guided Pruning and Foveated Rendering”
arXiv preprint arXiv:2407.00435, 2024 (preprint and shortened version of MetaSapiens)

PATENTS AND PROJECTS

- Dingsheng Luo, Xihong Wu, Yifan Yuan, Wenfei Hu, Weiming Qu, Yudi Zou, Jiawen Wang, Hongyu Fang, [Weikai Lin](#)
Control Method and System for Improving Operation Precision of Robot Arm
CN202310068025.4, China; published as CN116079730A, 2023 (**patent**)
- Dingsheng Luo, Xihong Wu, Shuai Fang, Jianan Zhang, [Weikai Lin](#), Tianlin Liu
Active Auditory Positioning Method for Map-Free Navigation
CN202210079214.7, China; published as CN114563011A, 2022 (**patent**)
- Dingsheng Luo, Xihong Wu, Jianan Zhang, Shuai Fang, Tianlin Liu, [Weikai Lin](#), Hongyu Fang
Robot Map-Free Navigation Method Based on Time Sequence Information Modeling
CN202110018866.5, China; published as CN112857370A, 2021 (**patent**)
- “Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality: An FPGA Demo”
Open-sourced project with optimized CPU/GPU/FPGA implementations. [\[code\]](#)

TEACHING EXPERIENCE

Teaching Assistant, University of Rochester

- Fall 2025, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2025, CSC 252/452 *Computer Organization*
- Fall 2024, CSC 257/457 *Computer Networks*

EXTERNAL SERVICE

Reviewer

- [ISMAR](#): 2026
- [SIGGRAPH](#): 2025, 2026
- [SIGGRAPH Asia](#): 2025
- [WACV](#): 2026
- [TCAD](#): 2025
- [JCST](#): 2026